

SOLAR SYSTEM GUIDE

Cheat sheet of fun facts, planet info, and other objects in our solar system.

THE SUN

A medium-sized G-type star providing nearly all energy in the Solar System.

- Makes up 99.86% of total Solar System mass
- Surface temperature ~5,500°C
- Core temperature ~15 million °C
- Solar wind forms a protective bubble called the heliosphere

INNER PLANETS

MERCURY

Smallest planet and closest to the Sun; rocky and cratered.

- No moons
- Very thin exosphere (not a real atmosphere)
- Huge temperature swings: -170°C to +430°C
- Slow rotation: 1 Mercury day = 176 Earth days
- Year length: 88 Earth days

VENUS

Earth-sized but extremely hostile; thick atmosphere traps heat.

- No moons
- Runaway greenhouse effect: hottest planet (~465°C)
- Rotates backward (retrograde) and very slowly
- Surface covered in volcanoes and lava plains
- Clouds made of sulfuric acid

EARTH

The only known world with stable liquid water and life.

- One moon
- 71% of surface is water
- Active plate tectonics reshape the crust
- Atmosphere: nitrogen + oxygen
- Moon stabilizes Earth's tilt - stable seasons

MARS

Cold desert world with signs of ancient water.

- Major moons: Phobos (falling inward), Deimos (stable)
- Home to Olympus Mons - largest volcano known
- Thin CO₂ atmosphere
- Seasonal dust storms
- Past rivers and lakes suggested by surface minerals

OUTER PLANETS

JUPITER

Largest planet; gas giant with powerful storms.

- Major moons: Io, Europa, Ganymede, Callisto
- Europa has an ocean beneath ice (life potential)
- Ganymede: largest moon, has a magnetic field
- Great Red Spot storm >300 years old
- Strong magnetic field and faint ring system

SATURN

Known for its stunning ring system made of ice and rock.

- Major moons: Titan (thick atmosphere), Enceladus (water jets)
- Lowest density of any planet - could float in water
- Rings extend thousands of km but are extremely thin
- Titan has methane lakes and weather cycles

URANUS

An ice giant tilted almost on its side.

- Major moons: Titania, Oberon, Miranda, Ariel, Umbriel
- Extreme seasons due to 98° tilt
- Atmosphere of methane: blue-green color
- Narrow, faint ring system
- Coldest planetary atmosphere in the Solar System

NEPTUNE

Windy, dynamic ice giant with active storms.

- Major moon: Triton - retrograde orbit and nitrogen geysers
- Fastest winds in the Solar System (~2,100 km/h)
- Dark storm systems that appear and disappear
- Methane-rich atmosphere causes deep blue color

DWARF PLANETS

PLUTO

- Major moon: Charon - acts almost like a double-planet
- Heart-shaped nitrogen ice plain (Tombaugh Regio)
- Thin atmosphere that freezes onto surface in winter

CERES

- Located in the Asteroid Belt
- Bright salt deposits in Occator Crater
- Possibly contains subsurface brine pockets

ERIS

- More massive than Pluto
- Has one moon: Dysnomia
- Extremely distant, with a very long orbit

SOLAR SYSTEM REGIONS

ASTEROID BELT

- Between Mars and Jupiter
- Rocky bodies spread far apart
- Largest member: Ceres

KUIPER BELT

- Beyond Neptune
- Icy objects including Pluto
- Source of short-period comets

OORT CLOUD

- Spherical, distant shell of icy bodies
- Hypothesized, not directly observed
- Source of long-period comets

HELIOSPHERE

- Bubble created by solar wind
- Boundary: heliopause - where solar wind meets interstellar space